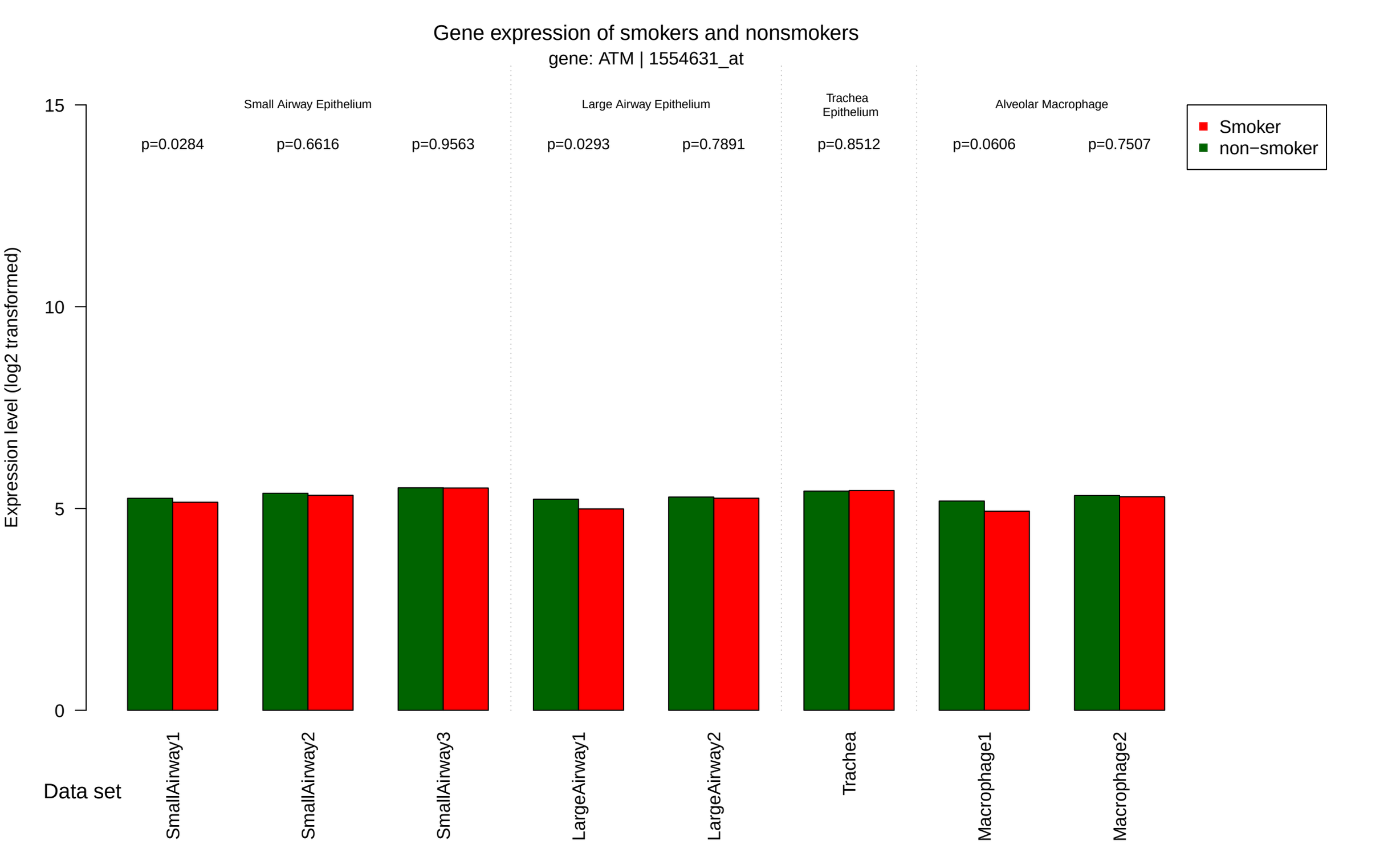
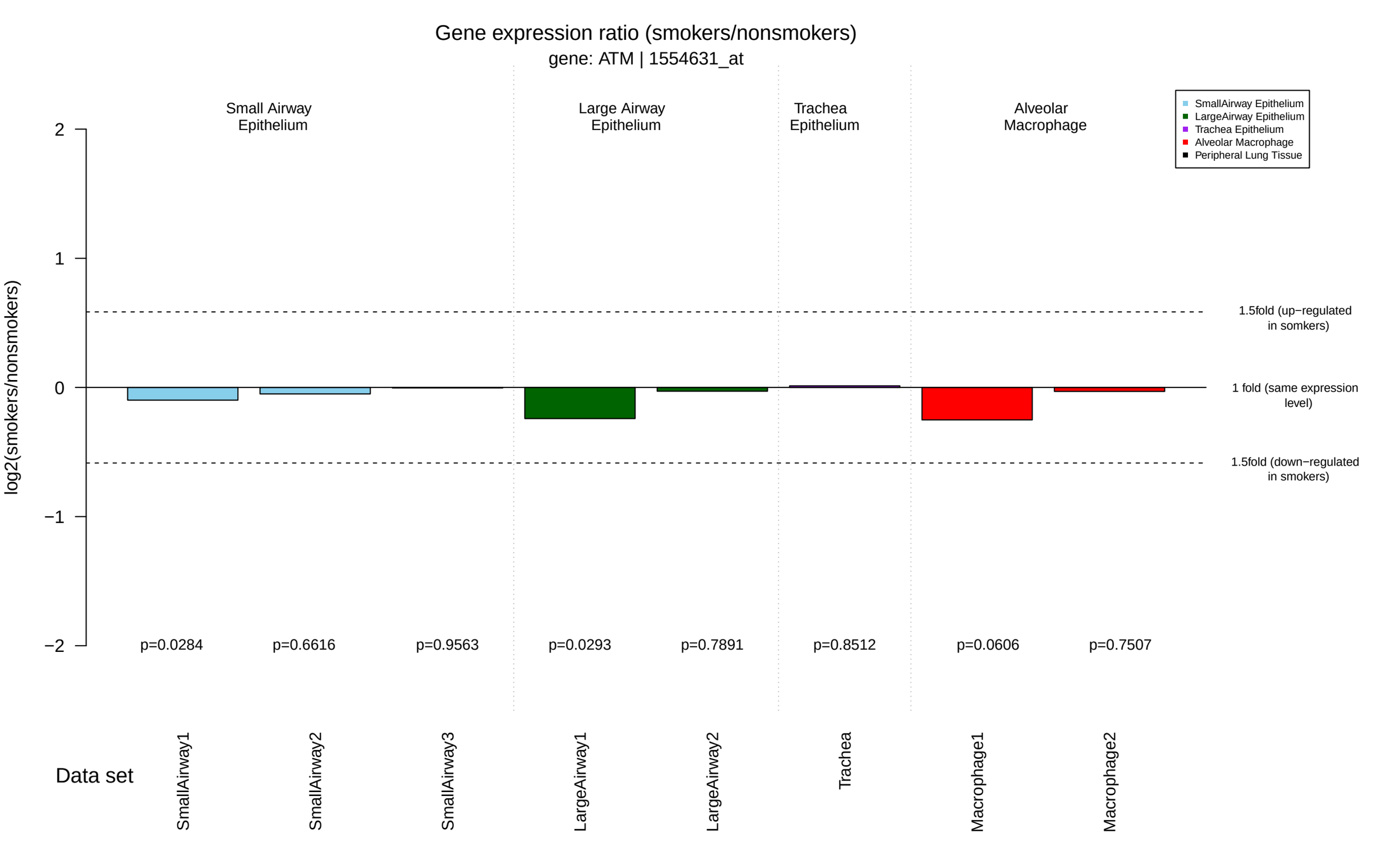
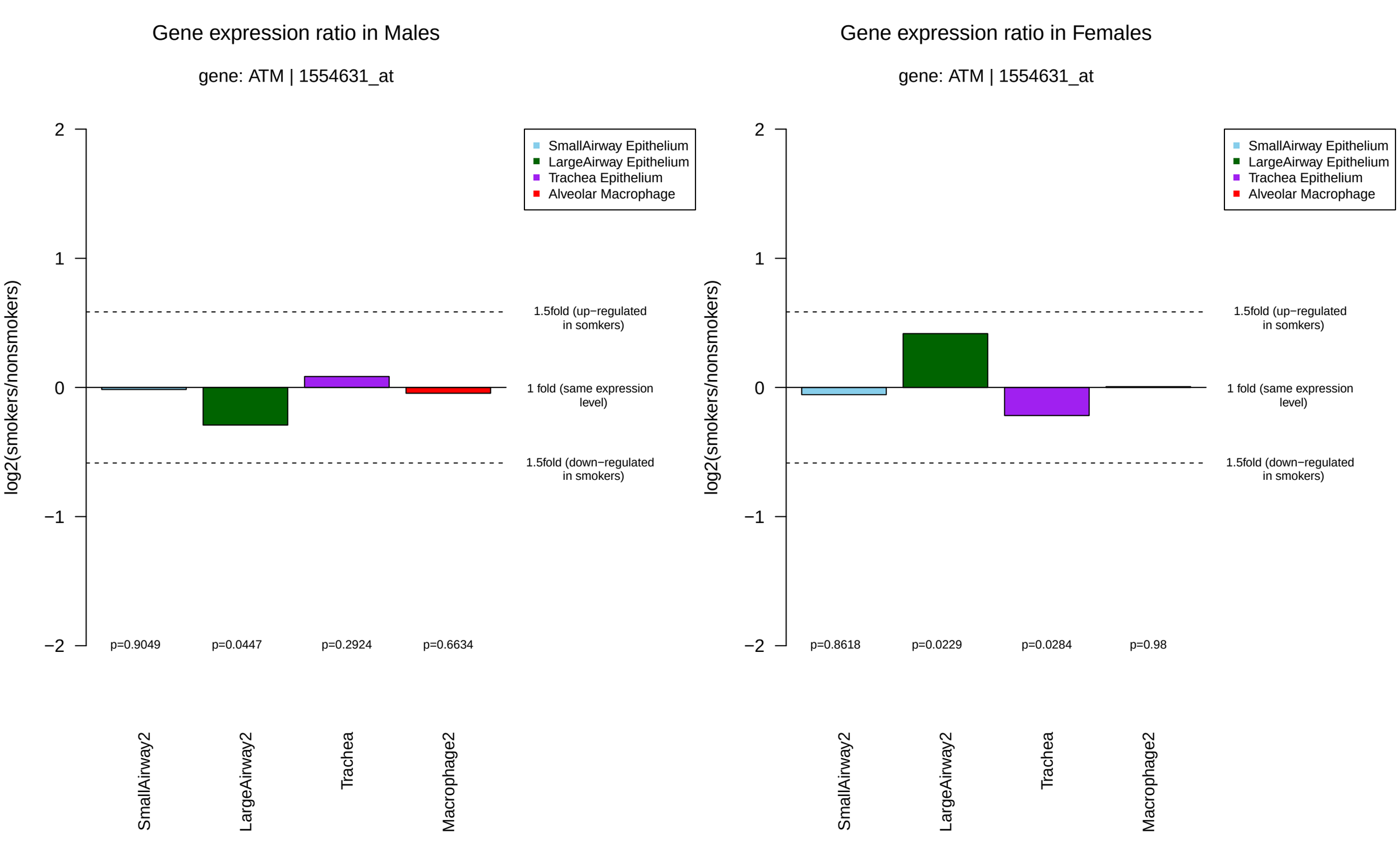
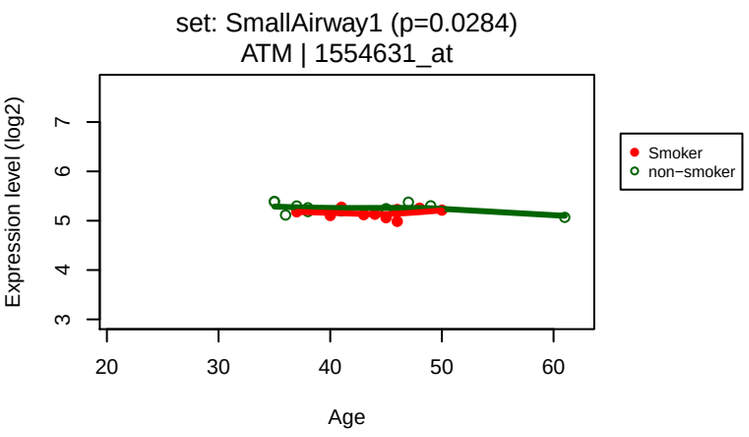
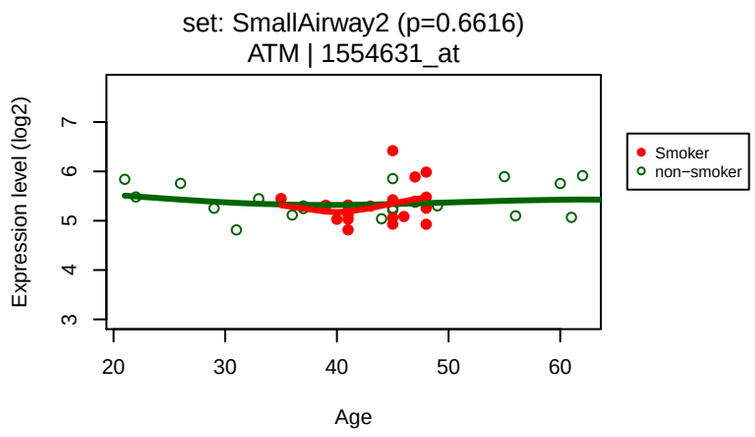
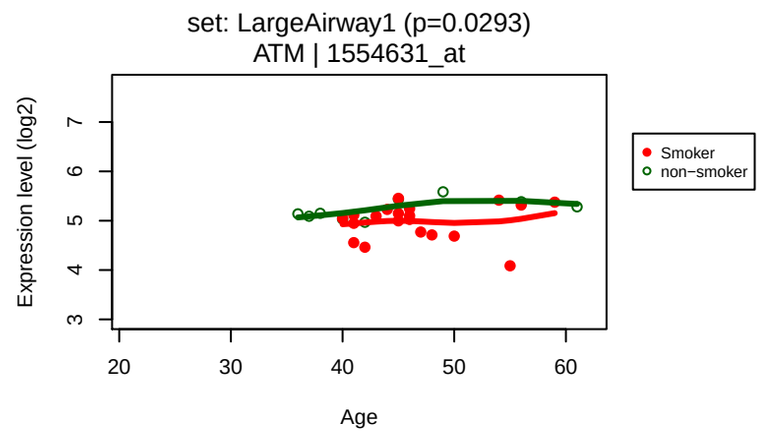

Gene expression of smokers and nonsmokers
gene: ATM | 1554631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0284
p=0.6616
p=0.9563
p=0.0293
p=0.7891
p=0.8512
p=0.0606
p=0.7507
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

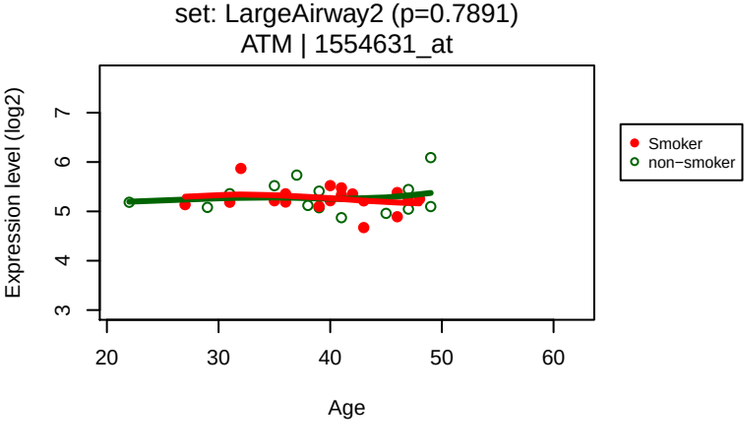

Gene expression ratio (smokers/nonsmokers)
gene: ATM | 1554631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0284
p=0.6616
p=0.9563
p=0.0293
p=0.7891
p=0.8512
p=0.0606
p=0.7507
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

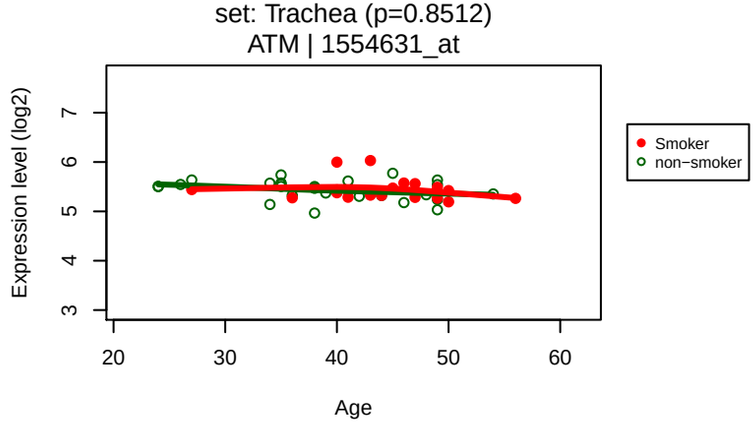

Gene expression ratio in Males
gene: ATM | 1554631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9049
p=0.0447
p=0.2924
p=0.6634
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATM | 1554631_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8618
p=0.0229
p=0.0284
p=0.98
SmallAirway2
LargeAirway2
Trachea
Macrophage2

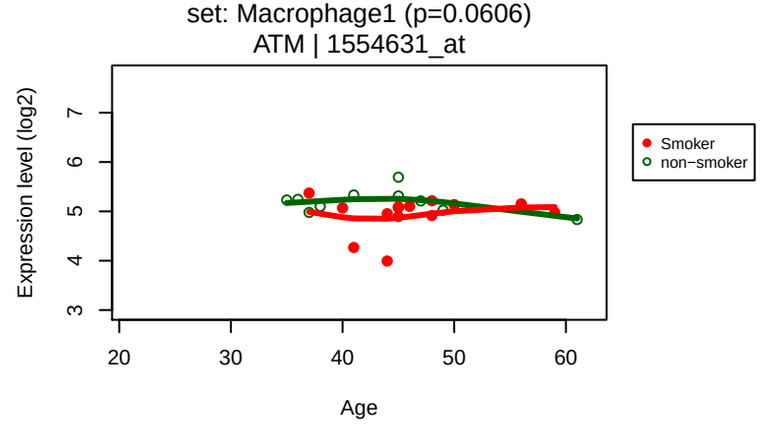

set: SmallAirway1 (p=0.0284)
ATM | 1554631_at
Expression level (log2)
Age
Smoker
non−smoker

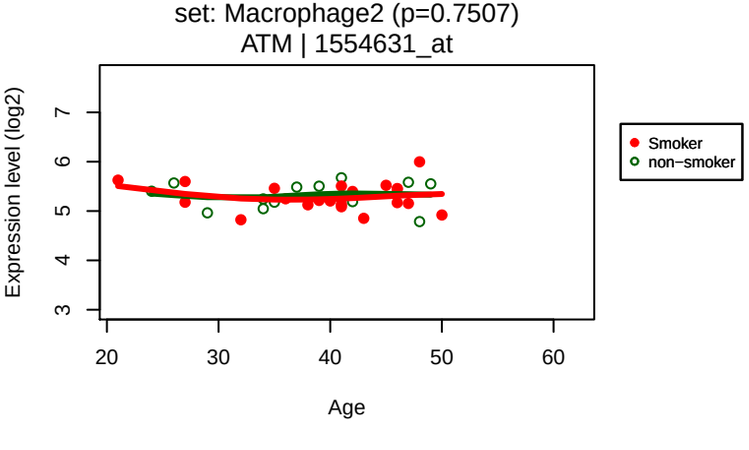

set: SmallAirway2 (p=0.6616)
ATM | 1554631_at
Expression level (log2)
Age
Smoker
non−smoker

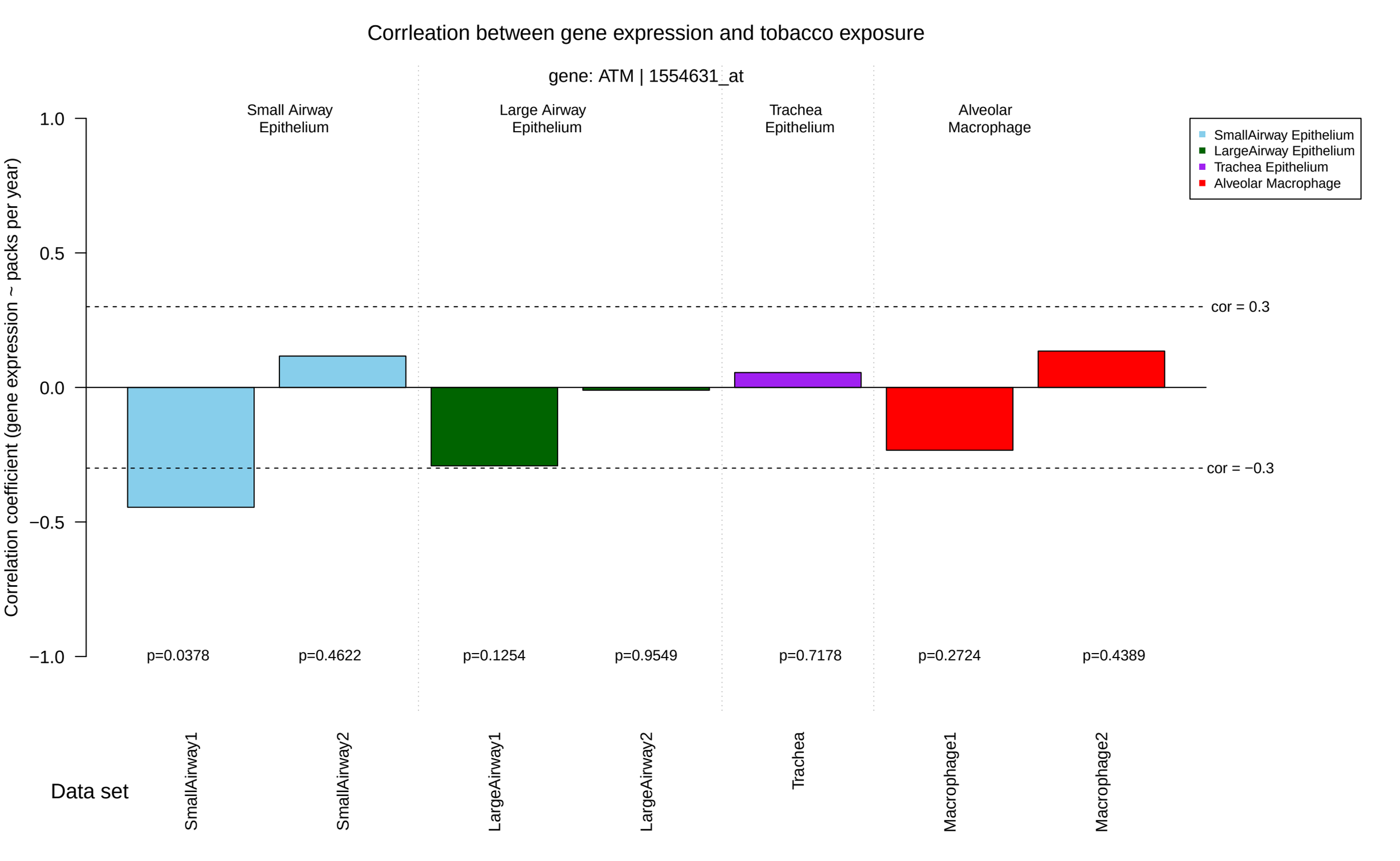

set: LargeAirway1 (p=0.0293)
ATM | 1554631_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7891)
ATM | 1554631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8512)
ATM | 1554631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0606)
ATM | 1554631_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7507)
ATM | 1554631_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATM | 1554631_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0378
p=0.4622
p=0.1254
p=0.9549
p=0.7178
p=0.2724
p=0.4389
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2